Roll No. | | | | | | | |

ODD END TERM EXAMINATION December - 2024

Name of the Course: **M.Tech** Semester: **Ist**

Name of the Paper: **Advanced Engineering Materials** Paper Code: **(DEME 21)**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	**(20 marks)**	**CO1, CO2, and CO5**
(a)	What are characteristics of glassy alloy materials? Explain with neat diagram and examples.	
(b)	Elaborate the metastable condition of alloys. Explain with key characteristics of metastable phase.	
(c)	Discuss the use of light weight alloys. What is the fabrication technique used for the development of light weight alloys?	
Q2	**(20 marks)**	**CO1, CO3, and CO4**
(a)	What are functional materials? Explain its objective of these materials near in future.	
(b)	Discuss brake-hardenable steel and elaborate the steps that are involved for the development of these materials.	
(c)	Explain the different categories of advanced high carbon steel with its microstructure, mechanical properties and application.	
Q3	**(20 marks)**	**CO1, CO2, and CO3**
(a)	Describe the advanced composite materials and how these materials are different from particulate materials?	
(b)	Elaborate the high temperature alloys and what is the basic difference from ultra-high temperature alloy?	
(c)	Discuss the principle of magneto-caloric effect with neat diagram and applictions.	
Q4	**(20 marks)**	**CO1, CO4, and CO5**
(a)	Why we use Rheocasting for the fabrication of automotive parts? Also, discuss the steps that are involved during fabrications.	
(b)	Explain the principle of Thixocasting Process and how it is different from rhoecasting?	
(c)	What is powder metallurgy route? Explain the main parameters that effect the strength of parts made from powder metallurgy route.	
Q5	**(20 marks)**	**CO1, CO2, and CO6**
(a)	Explain different types of 3D printing techniques used for the processing of advanced materials.	
(b)	Discuss the difference between rapid tooling, rapid manufacturing and rapid prototyping.	
(c)	What is the different non-destructive testing used for the estimation of defected parts?	